

Dimensions of Strange Attractors

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1 Dimensions of sets

Definition 1.1 (Box-counting dimension). *Let $Z \subset \mathbb{R}^m$ be bounded. Let $N(Z, \varepsilon)$ be the smallest number of balls of radius ε needed to cover Z . The lower and upper box dimensions are*

$$\underline{\dim}_B Z = \liminf_{\varepsilon \rightarrow 0} \frac{\log N(Z, \varepsilon)}{-\log \varepsilon}, \quad \overline{\dim}_B Z = \limsup_{\varepsilon \rightarrow 0} \frac{\log N(Z, \varepsilon)}{-\log \varepsilon}$$

If $\underline{\dim}_B Z = \overline{\dim}_B Z$, the common value is the box-counting dimension $\dim_B Z$.

Definition 1.2 (Hausdorff dimension). *Given $s \geq 0$, the s -dimensional Hausdorff measure is defined by*

$$\mathcal{H}^s(Z) = \liminf_{\delta \rightarrow 0} \left\{ \sum_i (\text{diam} U_i)^s : Z \subset \bigcup_i U_i, \text{diam} U_i < \delta \right\}$$

Hausdorff dimension is the critical value of s

$$\dim_H Z = \inf\{s : \mathcal{H}^s(Z) = 0\} = \sup\{s : \mathcal{H}^s(Z) = \infty\}$$

Hausdorff dimension is more rigorous and captures geometric scaling, but it is more difficult to estimate numerically. However, we can use the following inequality to bound it from above.

$$\dim_H Z \leq \underline{\dim}_B Z \leq \overline{\dim}_B Z$$

The issue with considering set dimensionality is that a subset of phase space only considers the attractor's support and not the underlying distribution of the orbits inside that set. This motivates the next chapter.

2 Dimensions of measures

The Hausdorff dimension of a measure μ can be defined by considering the set with the smallest dimensions that contains almost all of the mass.

Definition 2.1 (Hausdorff dimension of a measure). *Let μ be a finite Borel measure on a metric space X .*

$$\dim_H \mu = \inf\{\dim_H Z : \mu(X \setminus Z) = 0\}$$

There are several other dimensions of measures. The following two are particularly important for numerical approximations.

Definition 2.2 (Information dimension). *For a partition of space into boxes of size ε , let $p_i(\varepsilon)$ be the mass of the i -th box. The information dimension is morally*

$$D_1(\mu) = \lim_{\varepsilon \rightarrow 0} \frac{-\sum_i p_i(\varepsilon) \log p_i(\varepsilon)}{-\log \varepsilon},$$

when the limit exists.

Definition 2.3 (Correlation dimension).

$$C(r) = \int \mu(B(x, r)) d\mu(x), \quad D_2(\mu) = \lim_{r \rightarrow 0} \frac{\log C(r)}{\log r}.$$

For a finite orbit sample $x_1, \dots, x_N$, the following numerical approximation can be used.

$$C_N(r) = \frac{1}{N(N-1)} \#\{(i, j) : i \neq j, d(x_i, x_j) < r\}.$$

Correlation dimension counts pairs of points in an orbit that are closer than r to each other, which unlike the box-counting dimension, makes it dependent on the distribution of points in an orbit.

The notion of the dimension of a measure can be extended by considering the dimension of local neighbourhoods of points in X , which results in a curve of dimensions.

Definition 2.4 (Pointwise dimension). *The pointwise or local dimension of μ at x is*

$$d_\mu(x) = \lim_{r \rightarrow 0} \frac{\log \mu(B(x, r))}{\log r},$$

if the limit exists.

Intuitively, if $\mu(B(x, r)) \approx r^{d_\mu(x)}$ for small r , then the smaller the local dimension, the more concentrated the mass is near x .

Definition 2.5 (Exact dimensionality). *A measure is exact dimensional if there exists d such that*

$$d_\mu(x) = d$$

for μ -almost every x .

Theorem 2.6 (Young's dimension criterion). *Suppose μ is a exact dimensional finite measure on a metric space and for μ -almost every x . Then the natural dimensions of μ coincide and equal d .*

$$\dim_H \mu = \underline{\dim}_B \mu = \overline{\dim}_B \mu = d$$

3 Invariant, ergodic, and physical measures

Definition 3.1 (Invariant measure). *Let $(X, \mathcal{B})$ be a measurable space and $f : X \rightarrow X$ measurable. A probability measure μ is f -invariant if*

$$\mu(f^{-1}A) = \mu(A)$$

for every measurable set A

A dynamical system is called ergodic if every invariant measurable set has measure either 0 or 1.

Theorem 3.2 (Birkhoff ergodic theorem). *Let $f : X \rightarrow X$ be a measurable map such that μ is invariant with respect to f , and let $\varphi \in L^1(\mu)$. Then the time averages*

$$\frac{1}{n} \sum_{k=0}^{n-1} \varphi(f^k x)$$

converge for μ -almost every x . If μ is ergodic, then the limit is

$$\int \varphi d\mu$$

for μ -almost every x .

Birkhoff's ergodic theorem relates the average behaviour along a single trajectory to the space average of the whole system.

Definition 3.3 (Physical measure). *For a probability measure μ , define its basin by*

$$B(\mu) = \left\{ x : \frac{1}{n} \sum_{k=0}^{n-1} \varphi(f^k x) \rightarrow \int \varphi d\mu \text{ for } \varphi \in C^0(X) \right\}$$

A probability measure μ is called a physical measure if $B(\mu)$ has positive Lebesgue measure.

Physical measures ensure that they describe the behaviour from a positive-volume set of initial conditions, which means its physically observable.

4 SRB measures

SRB (Sinai–Ruelle–Bowen) measures are a class of invariant probability measures used to describe the statistical behaviour of chaotic systems with hyperbolic behaviour.

Consider the follow decomposition of a hyperbolic system on an invariant set.

$$T_x M = E_x^s \oplus E_x^u$$

where vectors in E_x^s contract under forward iteration and vectors in E_x^u expand under forward iteration. If the space is restricted to two dimensions $M = \mathbb{R}^2$, then $T_x M \cong \mathbb{R}^2$, hence

$$\mathbb{R}^2 = E_x^s \oplus E_x^u$$

An SRB measure is a physical and invariant measure that is absolutely continuous with respect to Lebesgue measure along unstable directions, while singular along stable directions.

These measures are important as they describe the statistical behaviour from a set of initial conditions with a positive Lebesgue measure. For many chaotic systems with strange attractors, trajectories starting from such initial conditions exhibit converging behaviours described by an SRB measure. This allows us to connect this concept with physical quantities such as entropy and Lyapunov exponents.

5 Lyapunov exponents

Definition 5.1 (Lyapunov exponents). *For a differentiable map $f : M \rightarrow M$, Lyapunov exponents represent the exponential growth rates of tangent vectors.*

$$\lambda(x, v) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|D_x f^n v\|,$$

when the limit exists.

Theorem 5.2 (Oseledets' theorem, schematic). *Let f preserve an ergodic probability measure μ , and assume suitable regularity conditions. Then for μ -almost every x , there exist Lyapunov exponents*

$$\lambda_1 > \lambda_2 > \cdots > \lambda_k$$

and an invariant decomposition

$$T_x M = E_1(x) \oplus \cdots \oplus E_k(x)$$

such that vectors in $E_i(x)$ grow or contract exponentially at rate λ_i :

$$\|Df^n(x)v\| \approx e^{\lambda_i n} \quad \text{for } v \in E_i(x)$$

Definition 5.3 (Kaplan–Yorke dimension). *Let the Lyapunov exponents be ordered*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m$$

Let j be the largest integer such that

$$\sum_{i=1}^j \lambda_i \geq 0$$

The Kaplan–Yorke dimension is

$$D_{KY} = j + \frac{\sum_{i=1}^j \lambda_i}{|\lambda_{j+1}|}$$

In a two-dimensional space with $\lambda_1 > 0 > \lambda_2$ this simplifies to the following.

$$D_{KY} = 1 + \frac{\lambda_1}{|\lambda_2|}$$

This can be understood as the expanding direction contributing one full dimension, since it is absolutely continuous in that subspace, while the stable direction produces a fractional part. This results in the dimension being the balance between stretches and contractions in the system. Unlike the other dimensions, this one is directly motivated by the dynamics and not the underlying geometric complexities. This concept can be generalised by considering the entropy of the system.

6 Entropy

Metric Entropy is the average rate at which the information about the system is lost under iteration, or equivalently, it is the measure of unpredictability of a dynamical system. For sufficiently smooth hyperbolic systems, the statistical complexity is directly related to dynamical instability through Pesin’s entropy formula.

Theorem 6.1 (Pesin entropy formula). *For sufficiently smooth hyperbolic systems and an SRB measure μ ,*

$$h_\mu(f) = \sum_{\lambda_i > 0} \lambda_i m_i,$$

where $h_\mu(f)$ is metric entropy, λ_i are Lyapunov exponents, and m_i are their multiplicities.

The above results above suggest a relation between the dimension of an invariant measure, the complexity of the system, and the Lyapunov exponents. In particular, the entropy is determined by the positive Lyapunov exponents.

Ledrappier–Young theory provides a more rigorous relation between entropy and dimensions. For an ergodic hyperbolic invariant measure, the dimension can be decomposed along invariant subspaces.

$$T_x M = E_1(x) \oplus \cdots \oplus E_k(x)$$

Each invariant subspace contributes a partial dimension δ_i , such that

$$\dim \mu = \sum_i \delta_i$$

These partial dimensions are determined by entropy levels in a given subspace divided by the corresponding Lyapunov exponent.

$$\delta_i = \frac{h_i - h_{i-1}}{\lambda_i},$$

where $h_i - h_{i-1}$ is the entropy contribution associated with the i -th Lyapunov direction.

This simplifies in the two-dimensional hyperbolic case, where

$$\mathbb{R}^2 = E_x^u \oplus E_x^s, \quad \lambda_u > 0 > \lambda_s,$$

the dimension decomposes into unstable and stable contributions,

$$\dim \mu = \delta_u + \delta_s,$$

where

$$\delta_u = \frac{h_\mu(f)}{\lambda_u}, \quad \delta_s = \frac{h_\mu(f)}{|\lambda_s|},$$

For an two-dimensional SRB measure, Pesin's formula gives $h_\mu(f) = \lambda_u$, which agrees with the Kaplan–Yorke dimension, since this implies that $\delta_u = 1$ and $\delta_s = \frac{\lambda_u}{|\lambda_s|}$. While in two dimensions the reduction is simple, in higher-dimensional systems the Ledrappier–Young theory provides a richer understanding of dimensions.

7 Numerical results

The numerical investigation of the Hénon map

$$x_{n+1} = 1 - ax_n^2 + y_n, \quad y_{n+1} = bx_n,$$

with the standard parameter values

$$a = 1.4, \quad b = 0.3.$$

7.1 Dimension comparison

Quantity	Estimate	95% bootstrap interval
λ_1	0.420	n/a
λ_2	-1.624	n/a
D_{KY}	1.259	n/a
D_{box}	1.245	[1.235, 1.256]
D_2	1.202	[1.196, 1.207]
D_1	1.229	[1.222, 1.238]

Table 1: Global dimensions for the Hénon map at $a = 1.4$, $b = 0.3$.

A bootstrap resampling method is used to produce the confidence intervals. Lyapunov exponents were estimated by iteratively applying the Jacobian of the Hénon map to an orthonormal basis and then applying the QR decomposition.

7.2 Generalised dimensions

Definition 7.1 (Generalised dimensions). *Let μ be a probability measure on $\mathbb{R}^m$. For $\varepsilon > 0$, partition $\mathbb{R}^m$ into boxes of side length ε , and let $p_i(\varepsilon)$ be the μ -mass of the i -th box. For $q \neq 1$, the generalised dimension of order q is*

$$D_q(\mu) = \lim_{\varepsilon \rightarrow 0} \frac{1}{q-1} \frac{\log \sum_i p_i(\varepsilon)^q}{\log \varepsilon},$$

whenever the limit exists.

For $q = 1$, it is the information dimension.

q	D_q	95% bootstrap interval
0	1.245	[1.234, 1.257]
0.5	1.241	[1.234, 1.249]
1	1.229	[1.222, 1.238]
1.5	1.213	[1.205, 1.223]
2	1.191	[1.179, 1.202]
3	1.123	[1.102, 1.144]
4	1.048	[1.013, 1.079]

Table 2: Generalised dimension estimates D_q .

Uneven mass concentration since for larger q , D_q gives more weight to regions with larger mass.

7.3 Local dimensions vs orbit length

N	mean	95% mean interval	std. dev.	IQR
50000	1.266	[1.244, 1.288]	0.248	0.296
100000	1.250	[1.232, 1.269]	0.214	0.258
300000	1.244	[1.229, 1.261]	0.191	0.214
1000000	1.247	[1.231, 1.262]	0.171	0.172

Table 3: Local dimension estimates for increasing orbit length.

The mean stays close to the global dimension estimates as N increases, while the spread decreases, which shows exact dimensionality of the Hénon map.

7.4 Random perturbations

The investigation considered a perturbed Hénon system to simulate noise.

$$x_{n+1} = f(x_n) + \sigma \zeta_n,$$

where ζ_n is sampled uniformly from the unit disk, while σ controls the amplitude of the noise. The experiment used $N = 100000$ and 500 local dimension estimates for each noise level. The results are shown in Table 4.

σ	mean	median	std. dev.	IQR
0	1.239	1.243	0.228	0.297
0.0001	1.245	1.230	0.244	0.319
0.0005	1.276	1.268	0.220	0.288
0.001	1.262	1.245	0.222	0.283
0.002	1.283	1.266	0.206	0.292
0.003	1.336	1.318	0.227	0.299
0.004	1.372	1.357	0.212	0.269
0.005	1.421	1.405	0.202	0.251

Table 4: Local dimension estimates under increasing noise for $N = 100000$.

Clear increasing trend. Useless result? Always converges to 2 but with different speed.